# Spatial Analysis Metrics Reference

spatial_analysis_tools.py | fly_ovaries_processing_pipeline

## Overview

This module quantifies the spatial organisation of VASA+ germ cells in 3D Drosophila ovary fluorescence images. All analysis is performed exclusively on the VASA label image (instance segmentation, uint16 or uint8 TIFF). No co-registration with TJ or other channels is required.

The pipeline detects cell-to-cell contacts from shared voxel faces, builds a weighted NetworkX contact graph, and derives a hierarchical set of metrics at three levels: individual cell, population (per image), and spatial statistics.

**Colour key:**

- Per-cell (cell_metrics DataFrame)
- Per-sample summary (summary dict / CSV)
- Graph control (computeGraphControl)
- Ripley / Geometric (computeRipleysK / computeGeometricMetrics)

## Input / Output

| Item | Description |
|---|---|
| Label image (input) | 3D ZYX uint array -- refined VASA instance segmentation (*_masks_refined.tif) |
| px_size (input) | [Z, Y, X] voxel size in micrometers. Default pipeline: [0.25, 0.14, 0.14] |
| cell_metrics (output) | pandas DataFrame -- one row per cell, all per-cell columns |
| G (output) | networkx.Graph -- weighted contact graph (edge weight = shared surface area in um2) |
| summary (output) | dict -- population-level scalars for the image |
| control (output) | dict -- graph topology null test results |
| ripley (output) | dict -- K/L function arrays and CSR envelope |
| geom (output) | dict -- geometric shape metrics |
| *_VASA_neighbor_metrics.csv | Per-cell metrics exported by exportVASANeighborMetrics() |
| *_VASA_neighbor_summary.csv | Per-sample summary exported by exportVASANeighborMetrics() |

## 1 analyzeVASANeighbors -- Per-Cell Metrics

| | |
|---|---|
| **analyzeVASANeighbors()** | Contact detection + graph construction + per-cell metrics |

**Key parameters:**

| | | |
|---|---|---|
| **noise_threshold** | *default=5* | Minimum number of shared voxel faces to recognise a VASA-VASA contact. Filters sub-pixel noise at cell boundaries. Increase for noisier segmentations. |
| **px_size** | *required* | Voxel size [Z, Y, X] in micrometers. All surface and volume outputs are in physical units only if provided. |

*Contact detection: the label image is padded by 1 voxel and shifted along each axis. Pairs of different labels at each face boundary are collected and grouped. Each face contributes a physical area = px_size[Y]*px_size[X] (Z-faces), px_size[Z]*px_size[X] (Y-faces), or px_size[Z]*px_size[Y] (X-faces). Pairs below noise_threshold faces are discarded before graph construction.*

### 1A Morphology (per cell)

#### Cell volume
*col: volume_um3*

**Level:** Per cell | analyzeVASANeighbors
**Units:** micrometers^3 (um3)
**Formula / method:** voxel count (area from regionprops) x voxel_volume (pz x py x px)
**Interpretation:** Physical volume of the segmented cell. Used to compute equivalent diameter, mean_volume_um3 (population), and R_eq (geometric metrics).

#### Equivalent sphere diameter
*col: equivalent_diameter_um*

**Level:** Per cell | analyzeVASANeighbors
**Units:** micrometers (um)
**Formula / method:** d = (6 * volume_um3 / pi)^(1/3) -- diameter of a sphere with the same volume
**Interpretation:** Size proxy independent of cell shape. Used to normalise Ripley's r axis (r / mean_diameter_um) for cross-sample comparison.

### 1B Contact Graph (per cell)

#### Degree (number of VASA neighbours)
*col: vasa_neighbors*

**Level:** Per cell | analyzeVASANeighbors
**Units:** integer
**Formula / method:** k_i = degree of node i in the VASA contact graph G
**Interpretation:** How many other VASA cells cell i physically touches (above noise_threshold). k=0: isolated cell; k=1-2: peripheral; high k: interior/hub cell. Population distribution described by mean_k and std_k in the summary.

#### Component (cluster) size
*col: cluster_size*

**Level:** Per cell | analyzeVASANeighbors
**Units:** integer (number of cells)
**Formula / method:** Size of the connected component to which cell i belongs in G
**Interpretation:** All cells in the same connected component share the same value. Isolated cells have cluster_size=1. The largest value across all cells equals lcc_size.

#### VASA shared surface area
*col: vasa_surface*

**Level:** Per cell | analyzeVASANeighbors
**Units:** um2
**Formula / method:** Sum of contact areas between cell i and all its VASA neighbours
**Interpretation:** Raw contact area. Increases with both number of neighbours and the extent of contact per neighbour. Use shared_boundary_ratio for a normalised version.

#### Non-VASA surface area
*col: non_vasa_surface*

**Level:** Per cell | analyzeVASANeighbors
**Units:** um2
**Formula / method:** Surface area of faces touching label 0 (background / non-VASA space)
**Interpretation:** Surface not touching another segmented VASA cell. Includes contact with TJ cells, extracellular matrix, tissue boundary, and imaging artefacts. Cannot distinguish between these -- it is the embedding surface.
**Note:** TJ cells are segmented in a separate channel without spatial co-registration; their contribution to this surface cannot be separated.

#### Total surface area
*col: total_surface*

**Level:** Per cell | analyzeVASANeighbors
**Units:** um2
**Formula / method:** total_surface = vasa_surface + non_vasa_surface
**Interpretation:** Approximate total surface area of the cell estimated from face counting. Not a marching-cubes surface (blocky voxel estimate). Use for ratios, not as an absolute morphometric.

#### Shared boundary ratio (SBR)
*col: shared_boundary_ratio*

**Level:** Per cell | analyzeVASANeighbors
**Units:** dimensionless [0, 1]
**Formula / method:** SBR_i = vasa_surface_i / total_surface_i
**Interpretation:** Fraction of cell surface in contact with other VASA cells. SBR ~ 0: isolated, mostly embedded in non-VASA tissue. SBR ~ 1: almost fully surrounded by VASA neighbours (deep interior cell). Population mean SBR reflects how tightly the cluster is packed.
**Note:** SBR does NOT measure homotypic vs. heterotypic mixing because the 'other side' is undefined (could be TJ, ECM, or background). It measures embedding depth.

#### Clustering coefficient (CC)
*col: clustering_coeff*

**Level:** Per cell | analyzeVASANeighbors
**Units:** dimensionless [0, 1]
**Formula / method:** CC_i = (triangles through i) / (possible triangles through i) = 2*t_i / (k_i*(k_i-1))
Computed by networkx.clustering(G, i). Undefined (set to 0) when k_i < 2.
**Interpretation:** Probability that two neighbours of cell i also touch each other. CC=0: none of the neighbours are mutually connected (star/chain motif). CC=1: all neighbours form a complete clique. High mean CC implies compact, locally dense packing.
**Note:** CC is mathematically undefined for cells with degree < 2 (no triangle is possible). NetworkX assigns CC=0 in this case. Use mean_CC_defined in the summary, which excludes these cells. frac_CC_undefined quantifies how many cells are affected.

## 2 analyzeVASANeighbors -- Population Summary Metrics

*All summary metrics are in the summary dict returned by analyzeVASANeighbors() and are exported to *_VASA_neighbor_summary.csv by exportVASANeighborMetrics(). One row per image.*

### 2A Cell count and size

#### Number of cells
*col: n_cells*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** integer
**Formula / method:** Count of unique non-zero labels in the refined label image
**Interpretation:** Total VASA cells after IQR outlier removal. Always report alongside all other metrics as a covariate (many metrics scale with N).

#### Mean cell volume
*col: mean_volume_um3*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** um3
**Formula / method:** mean(volume_um3) across all cells in the image
**Interpretation:** Average cell size. Used to compute R_eq (geometric metrics) and equivalent_diameter_um. Differences across conditions may indicate changes in cell size independent of packing.

#### Mean equivalent diameter
*col: mean_diameter_um*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** um
**Formula / method:** mean(equivalent_diameter_um) across all cells in the image
**Interpretation:** Mean cell diameter used to normalise the Ripley r axis. Typical value: 5-8 um for Drosophila germ cells at 0.14 um/px.

## 2B Connectivity and fragmentation

### Number of connected components

*col: n_components*

**Level:** Per sample | analyzeVASANeighbors summary

**Units:** integer

**Formula / method:** networkx.number_connected_components(G) -- isolated cells count as size-1 components

**Interpretation:** Number of distinct disconnected groups of VASA cells. 1 = fully connected cluster. >1 = satellite groups or isolated cells. Increases with fragmentation_index.

**Note:** n_clusters is kept as a backward-compatible alias for n_components.

### Largest connected component size (LCC)

*col: lcc_size*

**Level:** Per sample | analyzeVASANeighbors summary

**Units:** integer (cells)

**Formula / method:** Size of the largest connected component in G

**Interpretation:** Number of cells in the main cluster. lcc_size / n_cells = fraction of cells in the dominant cluster. Used by computeGraphControl for CPL.

### Fragmentation index (FI)

*col: fragmentation_index*

**Level:** Per sample | analyzeVASANeighbors summary

**Units:** dimensionless [0, 1)

**Formula / method:** FI = 1 - lcc_size / n_cells
Derived from Schurch et al. 2020 (PMID 32763154)

**Interpretation:** FI = 0: single connected cluster (all cells in LCC). FI -> 1: all cells isolated. FI = 0.036: 96.4% of cells are in the main cluster, 3.6% are in satellite groups. More sensitive to small isolated groups than n_components alone.

### Component size distribution

*col: comp_size_p25 / p50 / p75 / max*

**Level:** Per sample | exportVASANeighborMetrics (CSV only)

**Units:** integer (cells)

**Formula / method:** Percentiles of the list of all connected component sizes, sorted descending

**Interpretation:** Describes the size distribution of all clusters. p50 = median cluster size. max = lcc_size. A bimodal distribution (one large + many size-1) indicates a compact main cluster with isolated satellites.

**Note:** The raw component_sizes list is not saved to CSV. The p25/p50/p75/max columns are written instead to avoid variable-length columns.

### Number of isolated cells

*col: n_isolated_cells*

**Level:** Per sample | analyzeVASANeighbors summary

**Units:** integer

**Formula / method:** Count of cells with degree k = 0

**Interpretation:** Cells with no VASA-VASA contacts above noise_threshold. May be genuine outliers, segmentation fragments, or cells at the cluster periphery.

### Fraction isolated

*col: frac_isolated*

**Level:** Per sample | analyzeVASANeighbors summary

**Units:** dimensionless [0, 1]

**Formula / method:** frac_isolated = n_isolated_cells / n_cells

**Interpretation:** Normalised version of n_isolated_cells. Comparable across images with different N.

## 2C Degree distribution

### Mean degree
*col: mean_k*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** float
**Formula / method:** mean(vasa_neighbors) across all cells
**Interpretation:** Average number of VASA-VASA contacts per cell. Reflects packing density. Higher mean_k = cells are more embedded in the cluster. Typical range for a compact 3D cluster: 3-7.

### Standard deviation of degree
*col: std_k*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** float
**Formula / method:** std(vasa_neighbors) across all cells
**Interpretation:** Heterogeneity of connectivity. Low std_k: cells are uniformly connected (regular packing). High std_k: hub cells and peripheral cells coexist.

## 2D Clustering coefficient summary

### Mean CC (degree >= 2 only)
*col: mean_CC_defined*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** dimensionless [0, 1]
**Formula / method:** mean(clustering_coeff) restricted to cells where vasa_neighbors >= 2
**Interpretation:** The only meaningful population-level CC average. Cells with degree < 2 have CC=0 by convention (undefined), and including them would bias the mean downward in fragmented samples. High mean_CC_defined: local triangular motifs are common = compact packing.

### Fraction CC undefined
*col: frac_CC_undefined*

**Level:** Per sample | analyzeVASANeighbors summary
**Units:** dimensionless [0, 1]
**Formula / method:** fraction of cells with vasa_neighbors < 2
**Interpretation:** Fraction of cells for which CC is undefined (degree 0 or 1). High values indicate a fragmented or sparsely connected population. Report alongside mean_CC_defined to contextualise it.

## 3 computeGraphControl -- Graph Topology Null Test

| **computeGraphControl()** | Configuration model null hypothesis test: is the observed graph topology non-random? |
|---|---|

**Key parameters:**

| | | |
|---|---|---|
| **n_iterations** | *default=1000* | Number of random graph realisations. Use 200 for exploratory runs, 1000 for publication. p-value resolution = 1/n_iterations. |

*Null model: Molloy-Reed configuration model (networkx.configuration_model). Each random graph preserves the exact degree sequence of the observed graph but randomises which nodes are connected. Multi-edges and self-loops are removed after construction. CPL is always computed on the LCC of each random graph because the configuration model can produce disconnected graphs after edge removal.*

*p-values are one-sided. Higher CC = more clustered, so p(CC) = fraction of null >= observed. Lower CPL = more compact, so p(CPL) = fraction of null <= observed.*

*Warning: CPL computation is O(V x E) per graph. A warning is emitted if N > 300. Reduce n_iterations for large graphs.*

### Observed metrics

#### Mean clustering coefficient (observed)

*col: ctrl_mean_clustering*

| | |
|---|---|
| **Level:** | Per sample \| computeGraphControl observed |
| **Units:** | dimensionless [0, 1] |
| **Formula / method:** | networkx.average_clustering(G) -- includes all cells (degree < 2 get CC=0) |
| **Interpretation:** | Observed graph-level CC. Compare to the null distribution mean to assess whether local clustering exceeds random expectation. p(CC) << 0.05: significantly more triangular motifs than random. |

#### Average shortest path length (CPL)

*col: ctrl_cpl*

| | |
|---|---|
| **Level:** | Per sample \| computeGraphControl observed |
| **Units:** | hops (graph distance, not micrometers) |
| **Formula / method:** | networkx.average_shortest_path_length(G_LCC) -- computed on LCC only |
| **Interpretation:** | Mean number of contact-graph steps between any two cells in the LCC. p(CPL) = 1.0: all null graphs have shorter CPL than observed. This is expected for a physical 3D spatial graph -- random graphs have long-range shortcuts absent in a contact graph (lattice-like structure). |
| **Note:** | CPL is in graph hops, not physical distance. A spatial contact graph always has longer CPL than a random graph of the same degree because spatial proximity constrains connections. |

#### Normalised CPL

*col: ctrl_cpl_norm*

| | |
|---|---|
| **Level:** | Per sample \| computeGraphControl observed |
| **Units:** | dimensionless |
| **Formula / method:** | CPL_norm = CPL / ln(N_LCC) -- removes N-dependence |
| **Interpretation:** | Size-corrected path length for comparing across images with different N. For a random graph, CPL ~ ln(N)/ln(<k>), so CPL_norm ~ 1/ln(<k>). For a spatial lattice, CPL_norm is larger. |

### Small-world index (SW)

*col: ctrl_sw*

**Level:** Per sample | computeGraphControl observed

**Units:** dimensionless

**Formula / method:** SW = (CC_obs / CC_rand_mean) / (CPL_obs / CPL_rand_mean)
Based on Watts & Strogatz 1998 (PMID 9623998)

**Interpretation:** SW > 1: the graph has higher CC AND/OR lower CPL than random. SW >> 1 for a compact physical cell cluster: CC_obs >> CC_rand dominates. SW ~ 1: topology is indistinguishable from random at both scales. Observed values 7-10 for control VASA clusters confirm strong small-world character.

**Note:** SW is sensitive to the ratio of both metrics. Because CPL_obs > CPL_rand for spatial graphs, SW is driven almost entirely by the CC ratio. Do not interpret as evidence of 'shortcuts'; interpret as evidence of local compactness.

## p-values

### p-value: mean clustering coefficient

*col: ctrl_p_mean_clustering*

**Level:** Per sample | computeGraphControl p_values

**Units:** dimensionless [0, 1]

**Formula / method:** fraction of null graphs with CC >= observed CC (one-sided, upper tail)

**Interpretation:** p << 0.05: observed CC is significantly higher than random rewiring. p = 0.000 (floor at 1/n_iterations): all null graphs have lower CC.

### p-value: CPL

*col: ctrl_p_cpl*

**Level:** Per sample | computeGraphControl p_values

**Units:** dimensionless [0, 1]

**Formula / method:** fraction of null graphs with CPL <= observed CPL (one-sided, lower tail)

**Interpretation:** p = 1.0: all null CPL <= observed CPL -- the observed graph has LONGER paths than any random rewiring. This is expected for spatial contact graphs (lattice-like; see CPL interpretation above). NOT evidence of poor clustering.

### p-value: n_components

*col: ctrl_p_n_components*

**Level:** Per sample | computeGraphControl p_values

**Units:** dimensionless [0, 1]

**Formula / method:** fraction of null graphs with n_components >= observed n_components

**Interpretation:** p = 1.0: random graphs are more fragmented than observed (expected: compact cluster). p << 0.05: observed graph has MORE components than expected from the degree sequence alone -- genuine spatial fragmentation beyond what random wiring predicts.

## 4 computeRipleysK -- Spatial Point Statistics

| computeRipleysK() | 3D Ripley's K and L functions with Monte Carlo CSR confidence envelope |
|---|---|

**Key parameters:**

| | | |
|---|---|---|
| **n_simulations** | *default=100* | Number of CSR Monte Carlo realisations for the confidence envelope. Increase to 500-1000 for publication-quality envelopes. |
| **n_steps** | *default=50* | Number of r values evaluated. Increase for smoother curves. |
| **mean_diameter_um** | *optional* | Pass summary['mean_diameter_um'] to obtain r_normalized output for cross-sample comparison. |

*Ripley's K uses cell centroids (in physical micrometers) as a spatial point pattern. It does NOT use the contact graph. K(r) counts how many cell pairs have centroids within distance r, normalised by point density. The L-transform L(r) = (3K/(4pi))^(1/3) - r linearises K so that L=0 under CSR. The 95% Monte Carlo CSR envelope is computed by randomly placing N points uniformly in the bounding box N_simulations times.*

*Voxel anisotropy is corrected by converting centroid coordinates to physical micrometers before any distance computation.*

### Outputs

#### r (distance axis)

*col: r [array, not in CSV]*

**Level:** Per sample | computeRipleysK
**Units:** micrometers
**Formula / method:** linspace(0, r_max, n_steps) where r_max = min(vol_dims) / 4
**Interpretation:** Distance values at which K and L are evaluated. r_max is set to 1/4 of the smallest image dimension to avoid edge effects.

#### L_observed (L-function)

*col: L_observed [array, not in CSV]*

**Level:** Per sample | computeRipleysK
**Units:** micrometers
**Formula / method:** L(r) = (3 * K_observed(r) / (4*pi))^(1/3) - r
L(r) = 0 under CSR (complete spatial randomness)
**Interpretation:** L(r) > 0: cells are more clustered at scale r than CSR. L(r) above the 95% envelope: statistically significant clustering. The scale of the peak indicates the characteristic cluster size.

#### r_normalized (diameter-normalised distance)

*col: r_normalized [array, not in CSV]*

**Level:** Per sample | computeRipleysK
**Units:** mean cell diameters (dimensionless)
**Formula / method:** r_normalized = r / mean_diameter_um
mean_diameter_um from analyzeVASANeighbors summary
**Interpretation:** Converts the r axis from micrometers to units of mean cell diameter. Enables direct comparison of L(r) curves across samples with different cell sizes. r_norm = 1 corresponds to one mean cell diameter.
**Note:** Only computed if mean_diameter_um is passed to computeRipleysK(). Pass summary['mean_diameter_um'] from the same image.

#### CSR confidence envelope

*col: L_envelope_lo / L_envelope_hi [arrays]*

**Level:** Per sample | computeRipleysK
**Units:** micrometers
**Formula / method:** 2.5th and 97.5th percentiles of L(r) from n_simulations CSR point patterns with the same N placed uniformly at random in the bounding box
**Interpretation:** The 95% pointwise envelope under the null hypothesis of CSR. L_observed above L_envelope_hi: significant clustering at that scale. L_observed below L_envelope_lo: significant regularity (inhibition).

## 5 computeGeometricMetrics -- Domain Shape Metrics

| | |
|---|---|
| **computeGeometricMetrics()** | Radius of gyration + surface/volume ratio of the merged VASA binary domain |

*These metrics describe the SHAPE and EXTENT of the whole VASA cell population, not individual cells. Two complementary views:*
*(1) Centroid cloud geometry: radius of gyration Rg -- how spread are the cell centres?*
*(2) Domain surface geometry: S/V of the merged binary mask -- how complex is the boundary?*

*Both are normalised by R_eq = (3*N*V_mean/(4*pi))^(1/3), the radius of a sphere containing the same total VASA cell volume. This removes the N- and size-dependence so values are comparable across images.*

### Radius of gyration

#### Radius of gyration (Rg)

*col: geom_rg_um*

| | |
|---|---|
| **Level:** | Per sample \| computeGeometricMetrics |
| **Units:** | micrometers |
| **Formula / method:** | Rg = sqrt( mean( \|r_i - r_cm\|^2 ) )<br>r_i = centroid of cell i in physical um (ZYX); r_cm = mean centroid |
| **Interpretation:** | RMS distance of cell centroids from the cluster centre of mass. Larger Rg = more spread out cluster. Scales with N^(1/3) for a compact sphere, so always report alongside n_cells. |

#### Equivalent sphere radius (R_eq)

*col: geom_r_eq_um*

| | |
|---|---|
| **Level:** | Per sample \| computeGeometricMetrics |
| **Units:** | micrometers |
| **Formula / method:** | R_eq = (3 * N * V_mean / (4*pi))^(1/3)<br>Radius of a sphere with the same total VASA cell volume |
| **Interpretation:** | Reference size for normalisation. Does NOT depend on spatial arrangement -- it is a volume-based quantity derived from cell count and mean cell volume. Larger R_eq = more cells or larger cells. |

#### Normalised radius of gyration (Rg_norm)

*col: geom_rg_norm*

| | |
|---|---|
| **Level:** | Per sample \| computeGeometricMetrics |
| **Units:** | dimensionless |
| **Formula / method:** | Rg_norm = Rg / R_eq<br>Reference: uniform solid sphere -> Rg/R = sqrt(3/5) ~ 0.775 |
| **Interpretation:** | Rg_norm ~ 0.775: centroid cloud consistent with a compact sphere. Rg_norm > 0.775: cluster more spread than a sphere of the same total volume. Note: discrete cell packing adds ~10-15% above 0.775 even for a compact cluster, so the effective compact reference is ~ 0.85-0.90. Observed values 1.06-1.24 indicate a moderately elongated cluster consistent with the tubular germarium geometry. |

## Surface / volume ratio

### Domain surface / volume ratio (S/V)

*col: geom_sv_ratio*

**Level:** Per sample | computeGeometricMetrics

**Units:** um^-1

**Formula / method:** sv_ratio = S_domain / V_domain
S_domain: marching-cubes surface area of binary mask (label_image > 0) with px_size spacing
V_domain: N * mean_volume_um3 (total VASA cell volume)

**Interpretation:** For a sphere of radius R: S/V = 3/R. Larger S/V = more complex boundary relative to the contained volume. Sensitive to cluster shape, surface roughness, and internal gaps between cells.

### Normalised S/V (sv_ratio_norm)

*col: geom_sv_ratio_norm*

**Level:** Per sample | computeGeometricMetrics

**Units:** dimensionless

**Formula / method:** sv_ratio_norm = sv_ratio * R_eq / 3
Reference: perfect sphere -> sv_ratio_norm = 1

**Interpretation:** sv_ratio_norm = 1: domain surface equals that of an equivalent sphere. > 1: excess surface from irregular shape, cell-boundary roughness, or internal gaps. Observed values 3.8-4.9 in control data. These high values reflect that marching cubes resolves individual cell surfaces, not just the outer hull. Use for RELATIVE comparison between conditions, not as an absolute shape descriptor. A significant decrease across conditions would indicate increased surface sharing.

**Note:** S_domain is computed from the merged binary mask using skimage.measure.marching_cubes with the physical voxel spacing. Each cell's outer surface is included except where cells share voxel-face boundaries. Internal gaps between cells contribute extra surface.

## 6 Output Files -- Column Reference

### *_VASA_neighbor_metrics.csv (per-cell)

| Column | Units | Description |
|---|---|---|
| label | int | Cell instance ID from label image |
| centroid_z/y/x | vox | Cell centroid in voxel coordinates (ZYX) |
| volume_um3 | um3 | Cell volume in cubic micrometers |
| equivalent_diameter_um | um | Equivalent sphere diameter |
| vasa_neighbors | int | Degree: number of VASA-VASA contacts (k_i) |
| cluster_size | int | Size of connected component containing this cell |
| vasa_surface | um2 | Shared surface area with VASA neighbours |
| non_vasa_surface | um2 | Surface area not touching another VASA cell |
| total_surface | um2 | Total cell surface (vasa + non_vasa) |
| shared_boundary_ratio | [0,1] | Fraction of surface touching VASA cells (SBR) |
| clustering_coeff | [0,1] | Local clustering coefficient (0 if degree < 2) |
| dataset | str | Dataset identifier (added in pipeline notebook) |
| condition | str | Experimental condition (added in pipeline notebook) |

### *_VASA_neighbor_summary.csv (per sample)

| Column | Units | Description |
|---|---|---|
| n_cells | int | Total VASA cells |
| n_components | int | Number of connected components |
| lcc_size | int | Largest connected component size |
| fragmentation_index | [0,1) | FI = 1 - lcc_size/n_cells |
| mean_cluster_size | float | Mean component size (cells) |
| median_cluster_size | float | Median component size |
| n_isolated_cells | int | Cells with degree = 0 |
| frac_isolated | [0,1] | n_isolated / n_cells |
| mean_k | float | Mean degree |
| std_k | float | Std of degree distribution |
| mean_CC_defined | [0,1] | Mean CC for cells with degree >= 2 |
| frac_CC_undefined | [0,1] | Fraction of cells with degree < 2 |
| mean_volume_um3 | um3 | Mean cell volume |
| mean_diameter_um | um | Mean equivalent diameter |
| comp_size_p25/p50/p75 | int | Percentiles of component size distribution |
| comp_size_max | int | Largest component size (= lcc_size) |

## 7 Quick Interpretation Guide

*Use this table to assess what each result means for a VASA germ cell cluster. Compact, healthy germ-cell niche = right column. Dispersed / perturbed = left column.*

| Metric | Dispersed / fragmented | Compact / clustered |
|---|---|---|
| fragmentation_index | High (> 0.1) | Low (< 0.03) |
| n_components | Many (> 5) | 1 or 2 |
| mean_k | Low (< 3) | High (> 5) |
| shared_boundary_ratio | Low mean (< 0.2) | High mean (> 0.4) |
| mean_CC_defined | Low (< 0.3) | High (> 0.5) |
| ctrl_p_CC | High (CC not unusual) | Low ~ 0 (CC >> random) |
| ctrl_sw | ~ 1 (no small-world character) | >> 1 (compact topology) |
| L(r) vs envelope | Within or below CSR envelope | Above envelope at small r |
| geom_rg_norm | High (>> 1) | Closer to 0.85-0.90 |
| geom_sv_ratio_norm | High (excess surface) | Relatively lower |

*General rules:*

- *Always report n_cells as a covariate (most metrics scale with N).*
- *Use mean_CC_defined, NOT mean clustering_coeff, for CC summary.*
- *CPL p = 1.0 is expected for spatial graphs; it does NOT indicate poor clustering.*
- *S/V norm >> 1 is expected; use it for relative comparison between conditions only.*
- *Ripley's L and graph metrics are complementary: L uses centroids (continuous space); graph metrics use contact topology (discrete). Use both.*